Kristen McDaniel

Illinois Hospital CHNA and County IPLAN Comparison Report

1. Project Background and Purpose

- Compare hospital Community Health Needs Assessments (CHNAs) with county health department IPLAN priorities across Illinois.
- Examine whether hospital and county health priorities align and how those patterns relate to population demographics and healthcare resources.
- Identify potential geographic disparities and propose a future GIS tool to visualize the findings.

2. Work Completed: Data Collection and Preparation

- Compiled hospital CHNA priorities and county health department IPLAN priorities.
- Created separate datasets and standardized health priority categories to support comparisons.
- Collected available assessment information and documented missing or unavailable plans.
- Prepared the datasets for geographic matching and comparison by county.

3. Proposed Analysis: Comparing CHNA and IPLAN Priorities

- Match hospital CHNA priorities with the corresponding county IPLAN priorities.
- Identify shared priorities, priorities unique to hospitals, and priorities unique to health departments.
- Calculate the percentage of priorities that overlap between the two sources.
- Compare priority alignment across counties and examine differences among multiple hospitals within the same county.
- **Proposed calculations:**
    - County priority coverage: Percentage of county IPLAN priorities also identified in hospital CHNAs.
    - Hospital priority coverage: Percentage of hospital CHNA priorities also identified in the county IPLAN.
    - Priority overlap: Percentage of all distinct priorities identified by either source that appear in both.

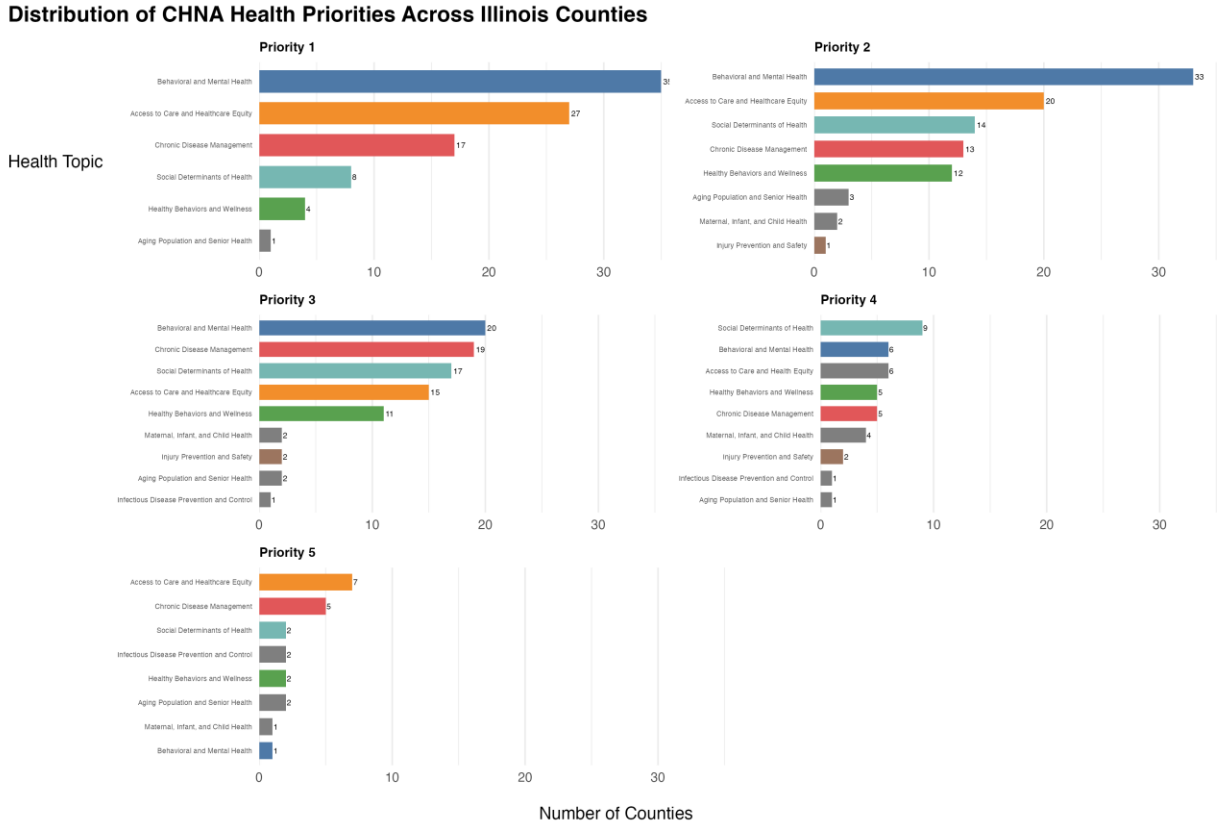
4. Proposed Population and Demographic Analysis

- Add U.S. Census Bureau population estimates and demographic data for each county.
- Include population size, poverty, and rurality.
- Compare population demographics with hospital availability and health priority patterns.

- Examine whether counties with greater socioeconomic disadvantage show different priority alignment or potential gaps in healthcare resources.
- **Proposed calculations**
    - Hospitals per 100,000 residents: Number of hospitals ÷ County population) × 100,000
    - Priority overlap percentage: (Number of shared priorities ÷ Total number of distinct priorities across both sources) × 100
    - Compare demographic percentages and hospital rates across counties to identify patterns that may warrant further investigation.

Note: Not all counties and hospitals had available data or completed Community Health Needs Assessments (CHNAs). Additionally, CHNA and IPLAN reporting years varied across jurisdictions, which may affect comparability. Therefore, the counts reflect the available data and should not be interpreted as a complete representation of all Illinois counties or hospitals.

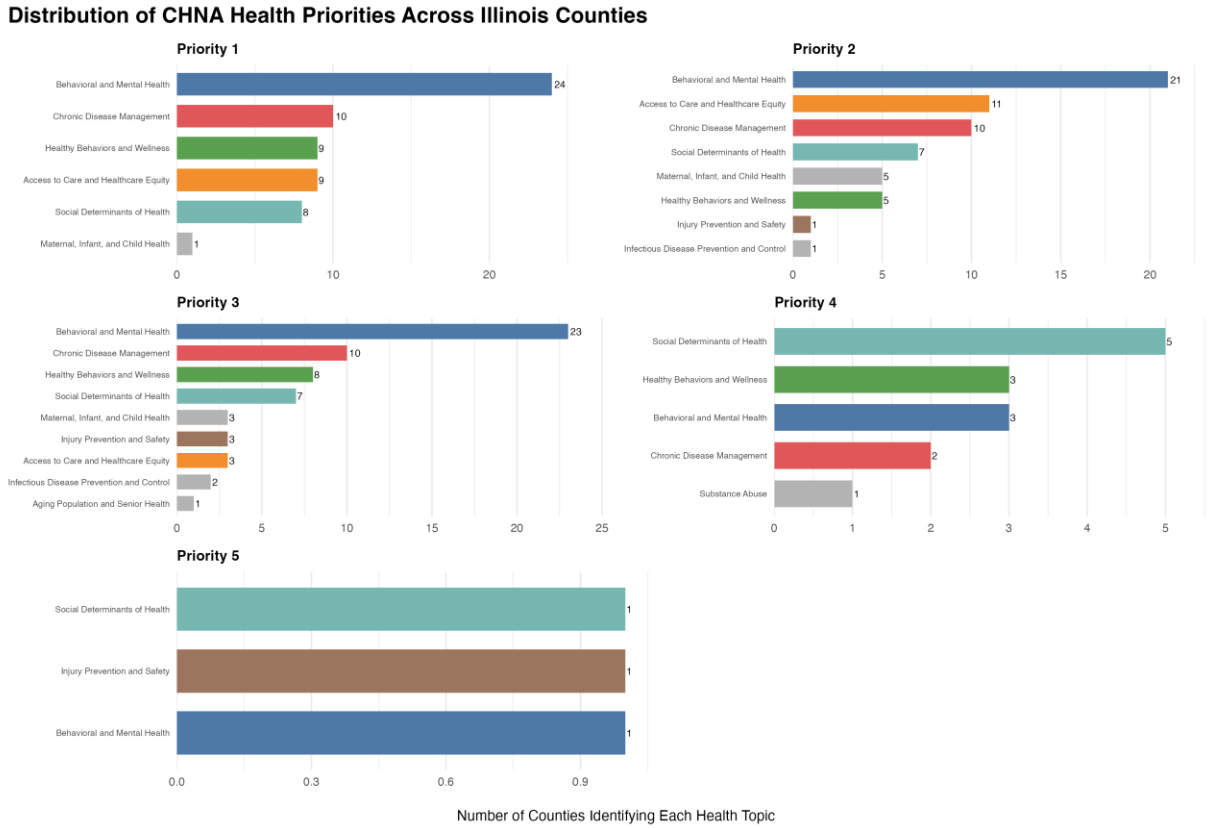
Distribution of Community Health Needs Assessment Health Priorities Across Illinois Hospitals



Hospital Community Health Needs Assessments (CHNAs) overwhelmingly rank Behavioral and Mental Health as Priority 1 and Priority 2. However, hospitals place a significantly higher initial priority on Access to Care and Healthcare Equity, making it the second most common Priority 1,

compared to county public health plans. This graph illustrates key institutional health system priorities; while hospitals align closely with counties on mental health needs, their strong initial emphasis on access to care reflects urgent clinical and operational challenges, such as emergency department utilization and insurance coverage barriers.

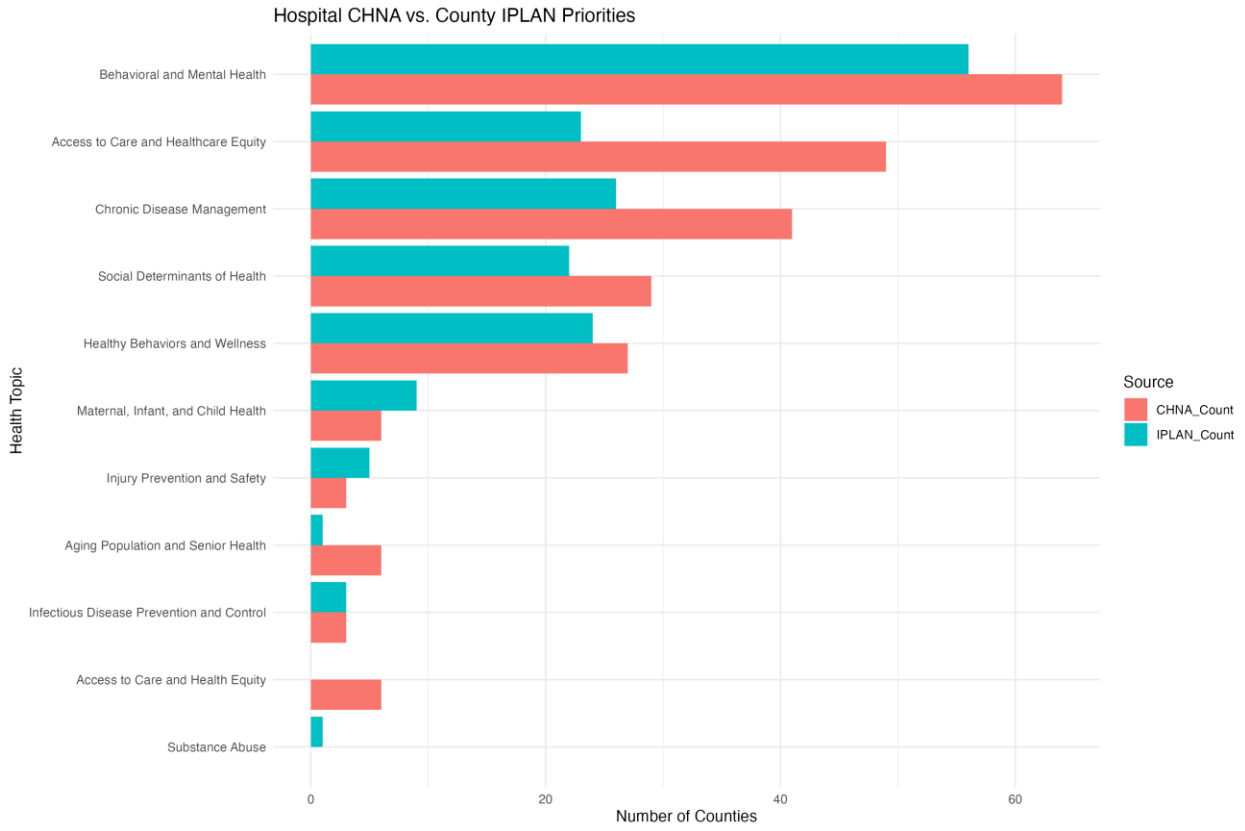
Distribution of Community Health Needs Assessment Health Priorities Across Illinois Counties



Across top-ranked priorities (Priority 1 through 3), Behavioral and Mental Health is consistently the most frequently identified health need by Illinois counties, followed by Chronic Disease Management, Healthy Behaviors and Wellness, and Access to Care and Healthcare Equity. Secondary and tertiary priorities (Priority 4 and 5) shift toward broader systemic factors like Social Determinants of Health. This visualization highlights a strong public health consensus across Illinois, demonstrating that mental health and chronic disease management form the primary core of county health assessment strategies statewide.

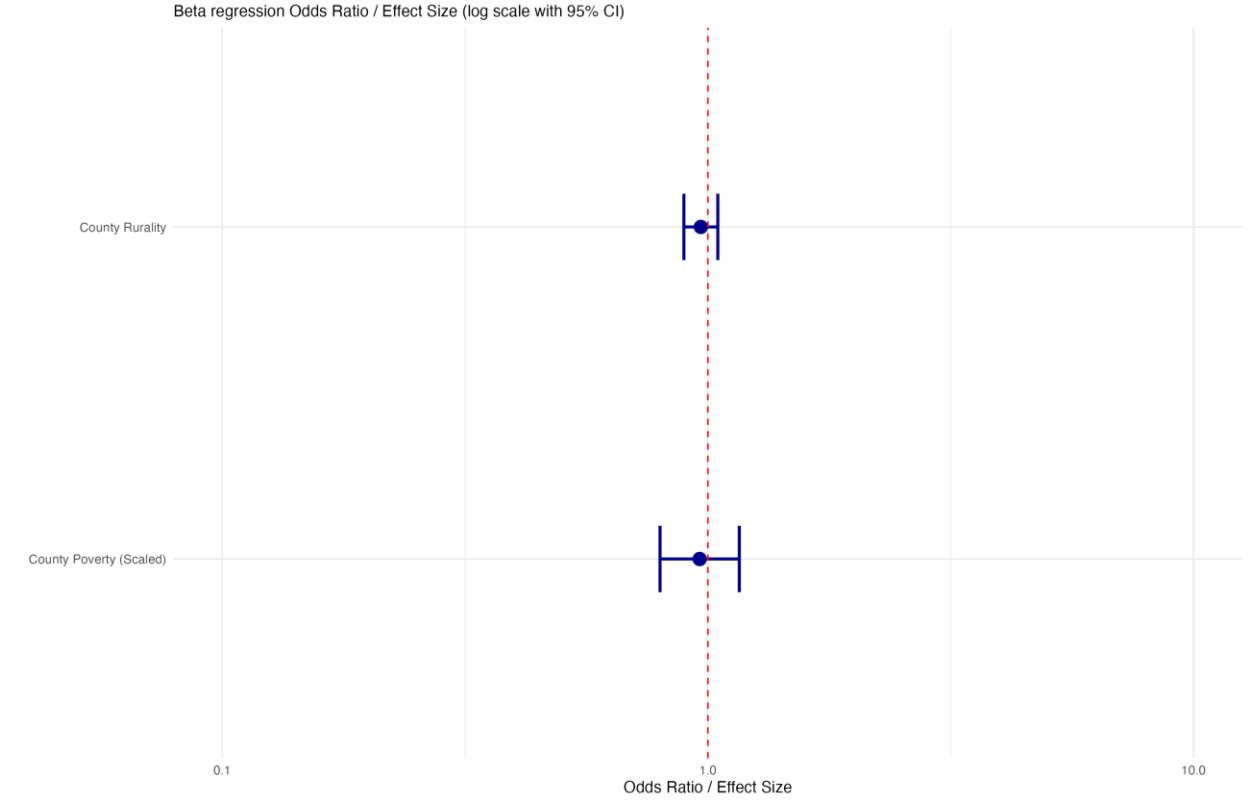
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Hospital Community Health Needs Assessment Health Priorities vs. County Illinois Project for Local Assessment of Needs Health Priorities



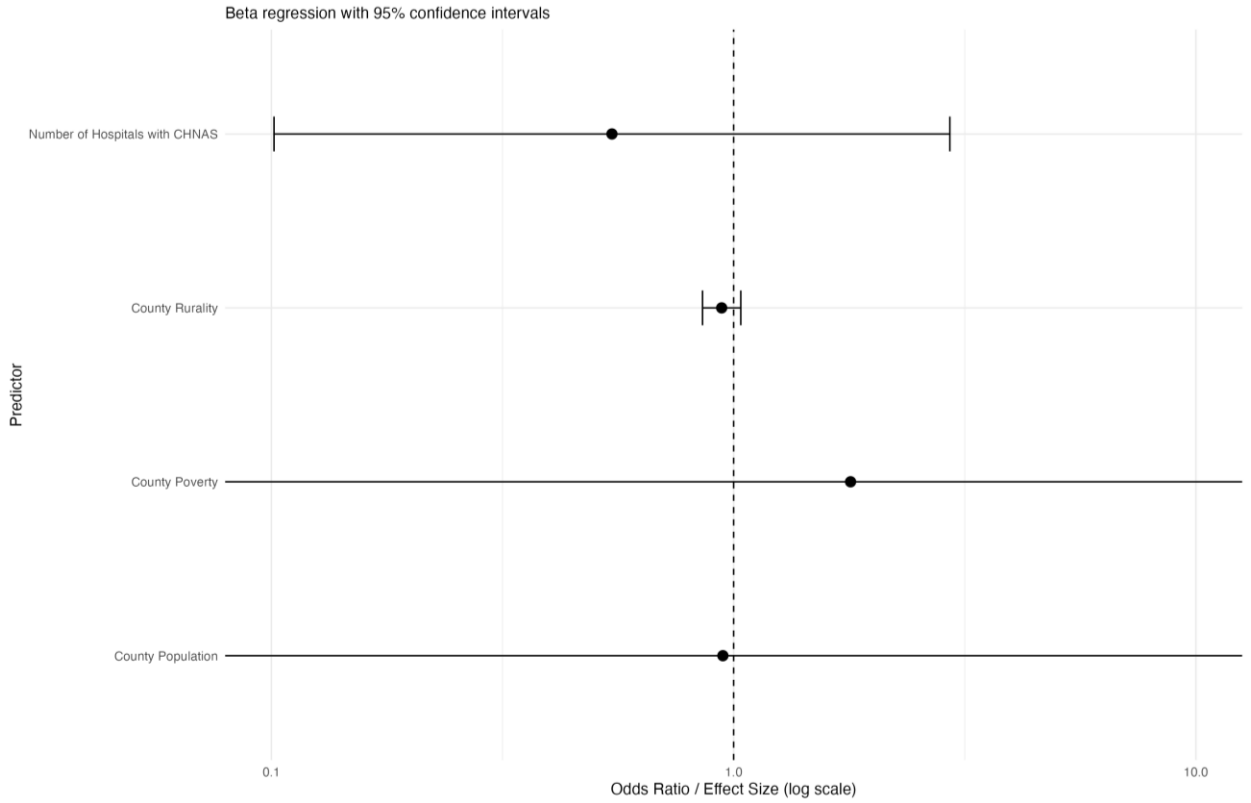
Hospital CHNAs identified Behavioral and Mental Health. as the most common, priority, followed by Access to Care and Healthcare Equity, and Chronic Disease Management. These topics were also prominent in country IPLAN priorities, although they were reported less frequently than in hospitals CHNAs. Social Determinants of Health and Healthy Behaviors and Wellness were also frequently identified by both sources. Overall, the findings suggest substantial overlap in major community health concerns, particularly behavioral health and healthcare access, while differences in counts may reflect differing priorities across hospital service areas and counties.

Factors Associated with County Priority Overlap



A refined beta regression specifically evaluating County Rurality and Scaled County Poverty shows point estimates for both variables centered almost directly at 1.0, with narrow confidence intervals clearly spanning the null line. This graph statistically confirms that neither rurality nor poverty scale directly influences the degree of priority overlap between entities. Consequently, priority alignment is shown to be independent of these demographic metrics, emphasizing that alignment depends on context-specific strategies rather than county demographics.

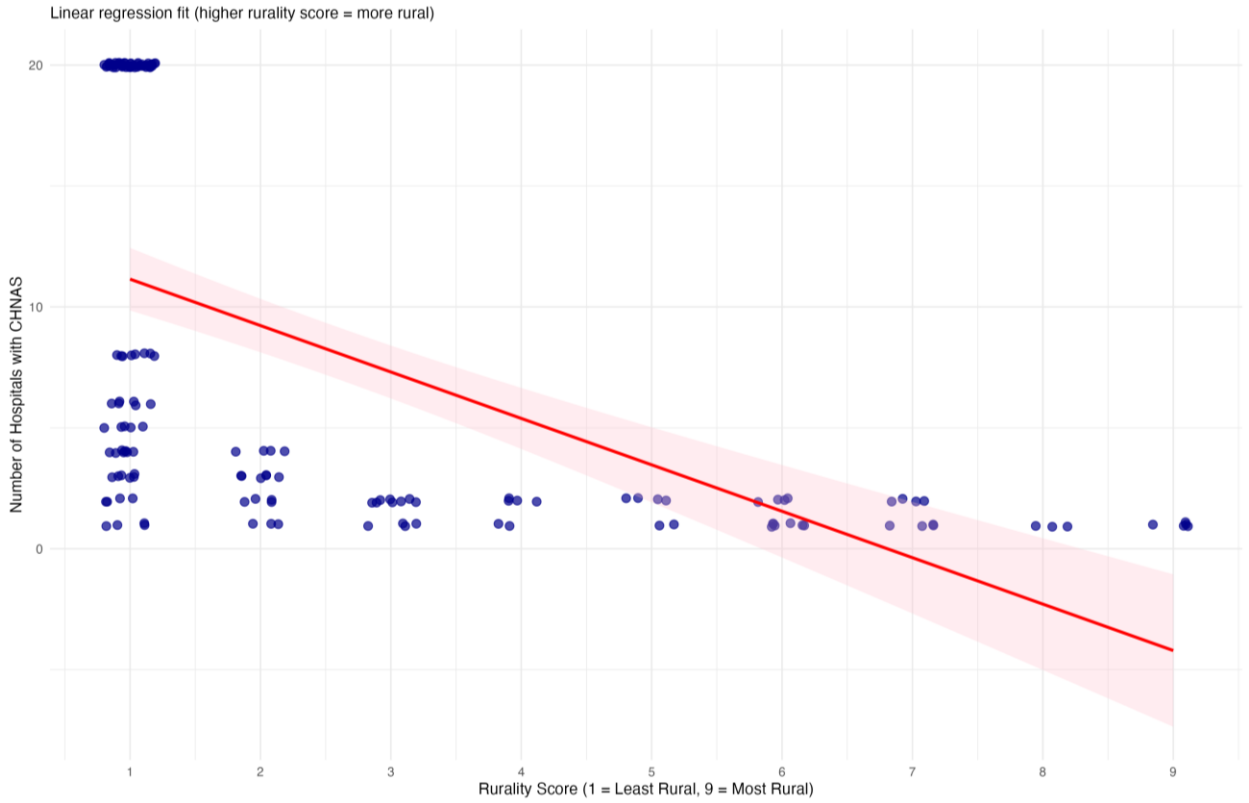
Factors Associated with County–Hospital Priority Overlap



Results from a beta regression examining the alignment between county and hospital priorities show that county poverty has a positive point estimate toward greater overlap, whereas county rurality and the number of hospitals lie slightly below 1.0. However, the 95% confidence intervals across all predictors are exceptionally wide and cross 1.0. This suggests that broad structural county characteristics like poverty or rurality do not reliably predict how closely hospital and county health priorities align, pointing instead toward localized collaboration dynamics as the key driver of alignment.

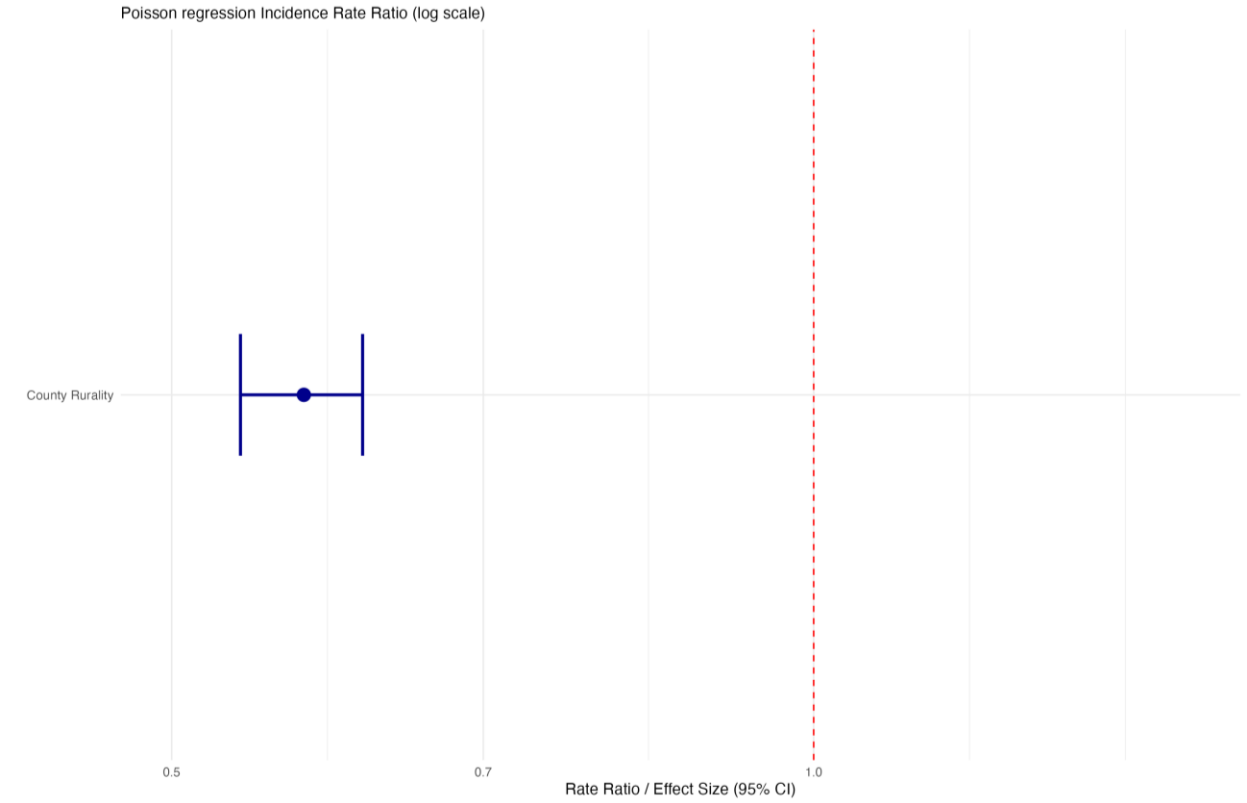
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Association Between County Rurality and the Number of Hospitals with CHNAs



The fitted line indicates a negative relationship between rurality score and the number of hospitals with CHNAs. Counties classified as more rural generally have fewer hospitals with CHNAs than less-rural counties. This pattern may reflect differences in population concentration and healthcare infrastructure. However, the distribution includes several observations with 20 hospitals, so the linear trend should be interpreted cautiously. The number of hospitals alone also does not measure hospital availability relative to population size or geographic access.

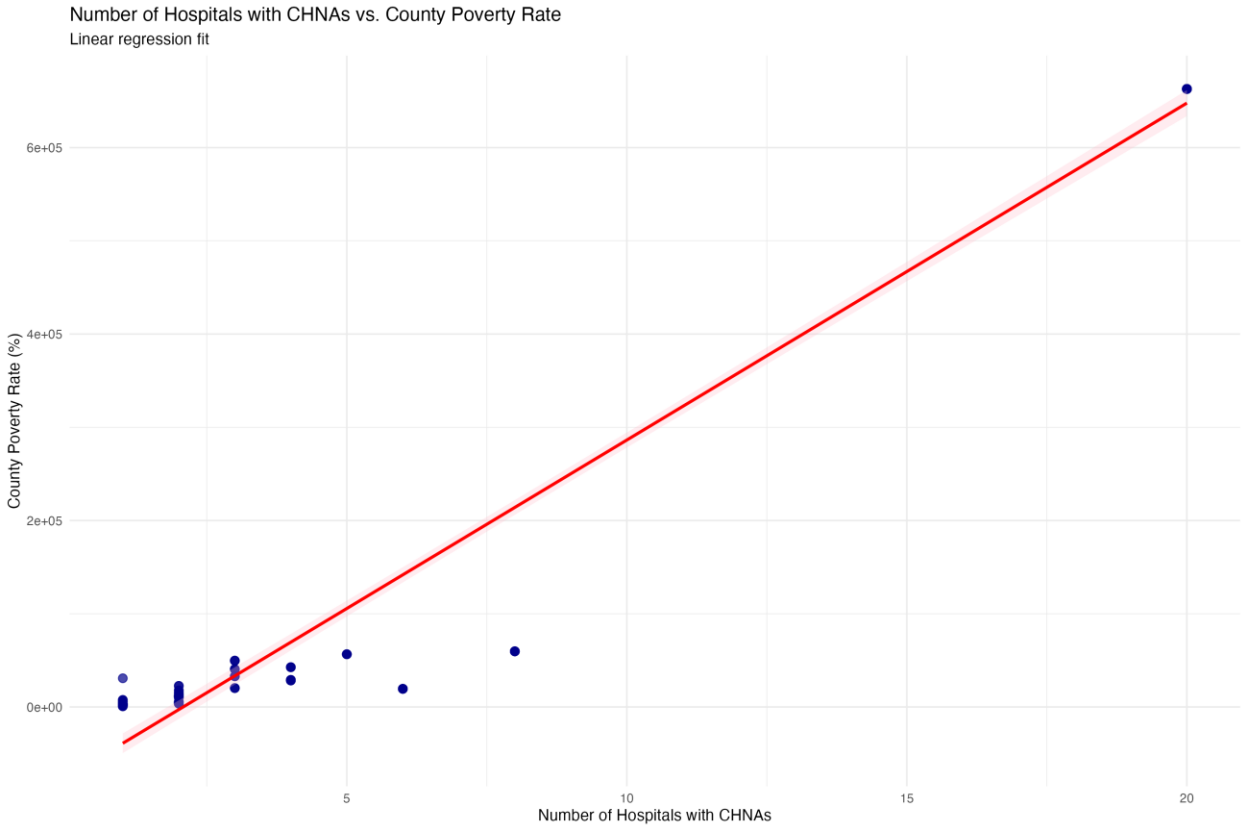
Association Between County Rurality and Hospital Count: Poisson Regression Estimates



A Poisson regression incidence rate ratio (IRR) plot shows that County Rurality has an effect size significantly below 1.0 (IRR 0.6), with its entire 95% confidence interval falling well to the left of the reference line. This graph statistically establishes a robust negative association between rurality and hospital density. Because more rural counties have significantly fewer hospitals per capita and geographic area, this finding highlights a key structural access barrier that complicates cross-sector health planning and resource allocation.

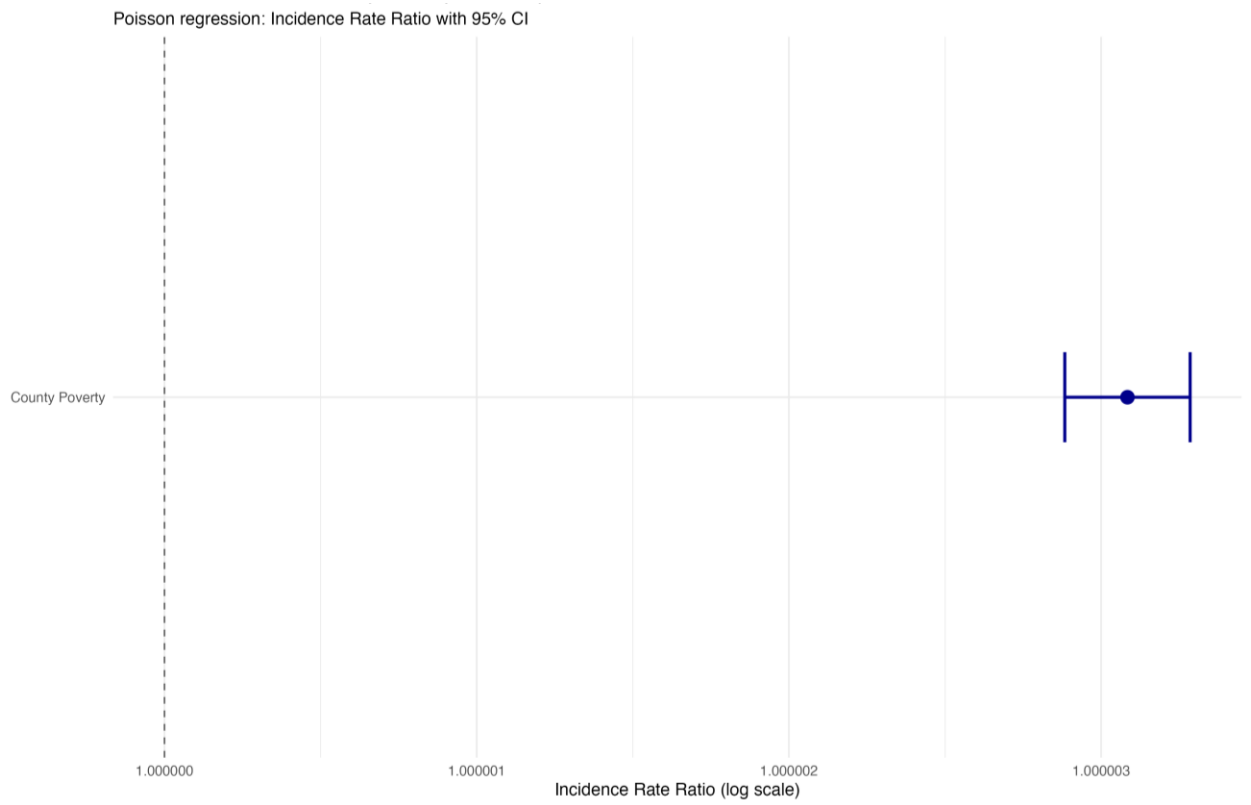
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Association Between Hospital Count and County Poverty Rate in Illinois



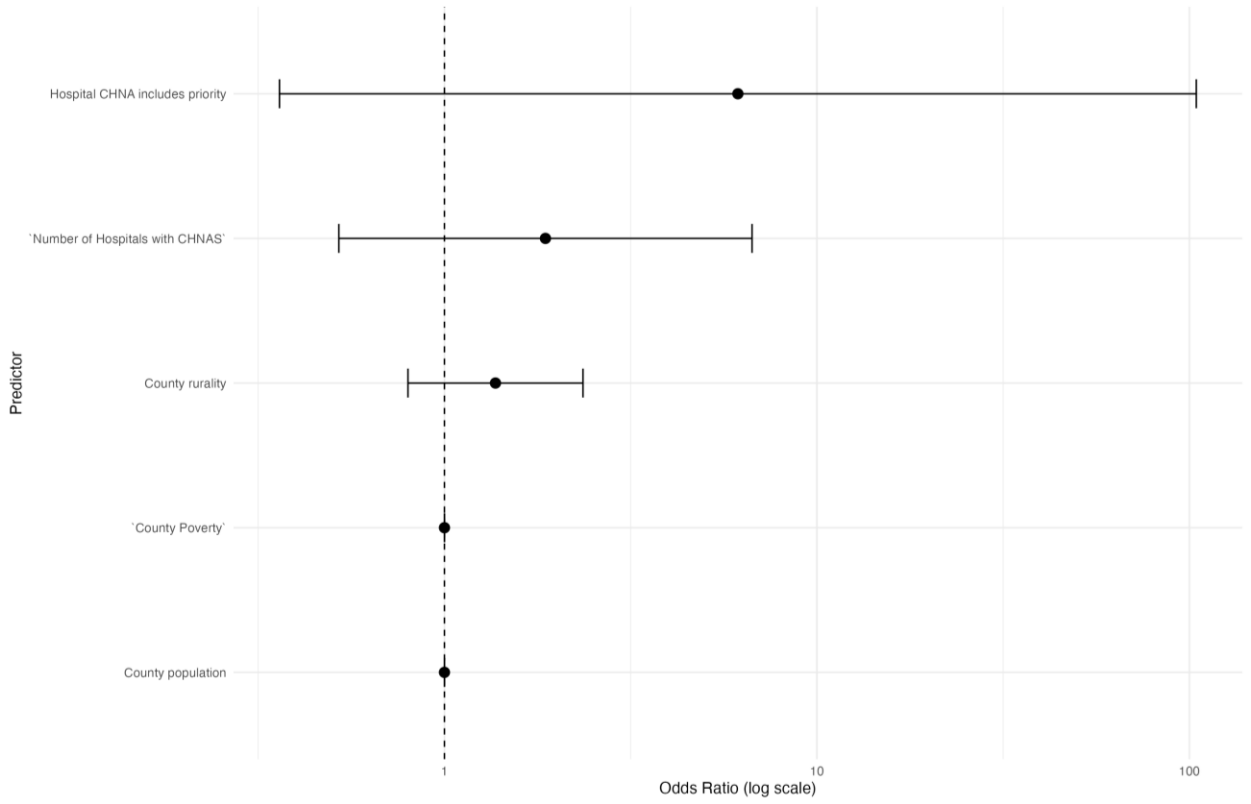
The plot shows a positive linear trend between the number of hospitals with available Community Health Needs Assessments (CHNAs) and county poverty rate. However, the apparent relationship is heavily influenced by an extreme observation at 20 hospitals and a poverty value that is far outside the scale expected for a percentage. This suggests a potential data or variable-labeling issue. The model should not be interpreted until the poverty variable is verified and the influential observation is investigated.

Association Between County Poverty and Hospital Count: Poisson Regression Estimates



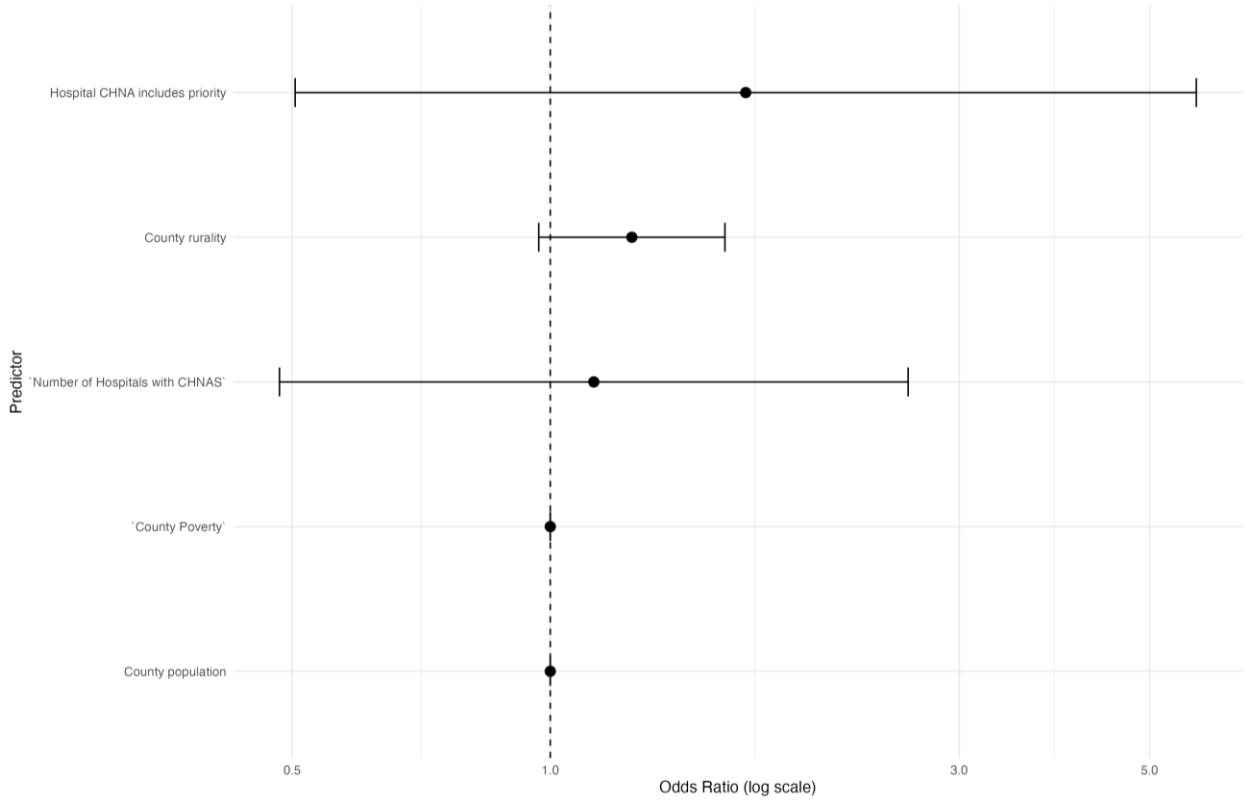
The estimated incidence rate ratio appears extremely close to 1, suggesting little estimated change in hospital count for a one-unit increase in the poverty predictor. However, the horizontal axis spans only a very narrow range around 1, making the effect difficult to interpret meaningfully. The poverty variable and its units should be verified, and the model should be refitted if the outcome or predictor was incorrectly specified. The current figure does not support a reliable conclusion about the relationship between poverty and hospital availability.

Factors Associated with Behavioral and Mental Health Priorities



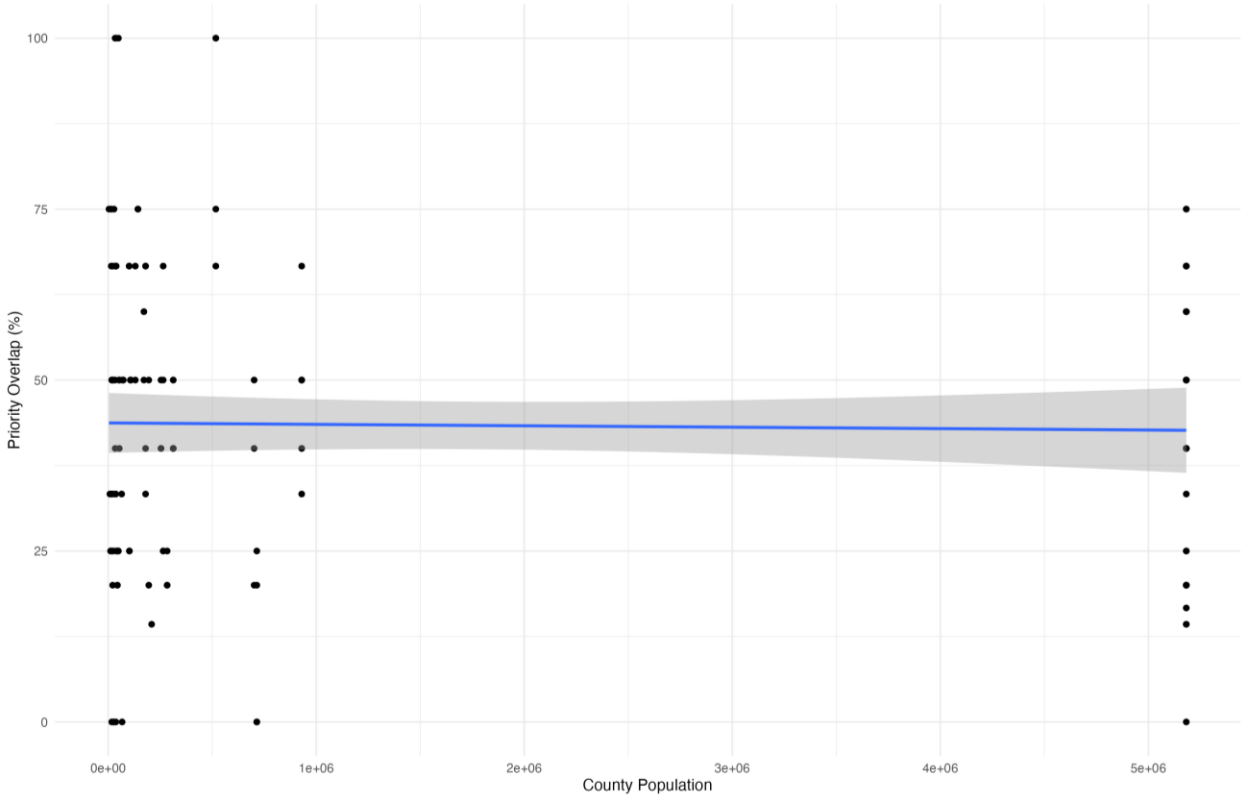
This forest plot displays odds ratios on a log scale to show factors predicting whether behavioral health is prioritized. The strongest positive predictor is whether local hospital CHNAs include it as a priority. County rurality and the number of hospitals with CHNAs also trend positive, though their 95% confidence intervals cross or approach 1.0, while county poverty and population show odds ratios centered at 1.0. These findings reveal that institutional alignment—specifically a local hospital's focus—is the strongest driver of whether a county formally prioritizes behavioral health, far outweighing baseline demographic factors like population size or poverty.

Factors Associated with Chronic Disease Management Priorities



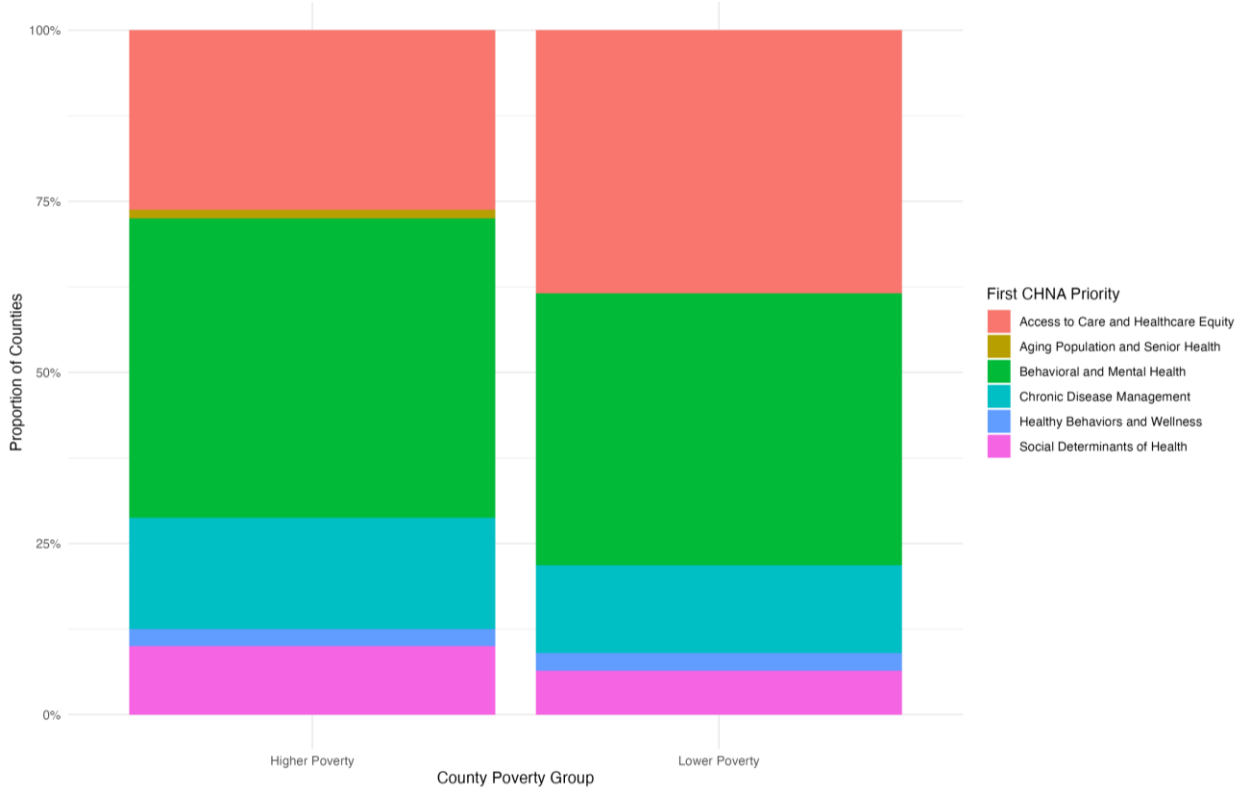
In evaluating chronic disease management, inclusion in a hospital CHNA demonstrates a strong positive point estimate, albeit with wide confidence intervals. County rurality displays a modest positive association, whereas the number of hospitals, county poverty, and county population all center around the null reference line of 1.0. This indicates that local hospital prioritization remains an influential factor in driving county-level chronic disease initiatives, while broader sociodemographic predictors alone do not strongly dictate whether chronic disease becomes a designated public health priority.

Relationship Between County Population Size and CHNA–IPLAN Priority Overlap



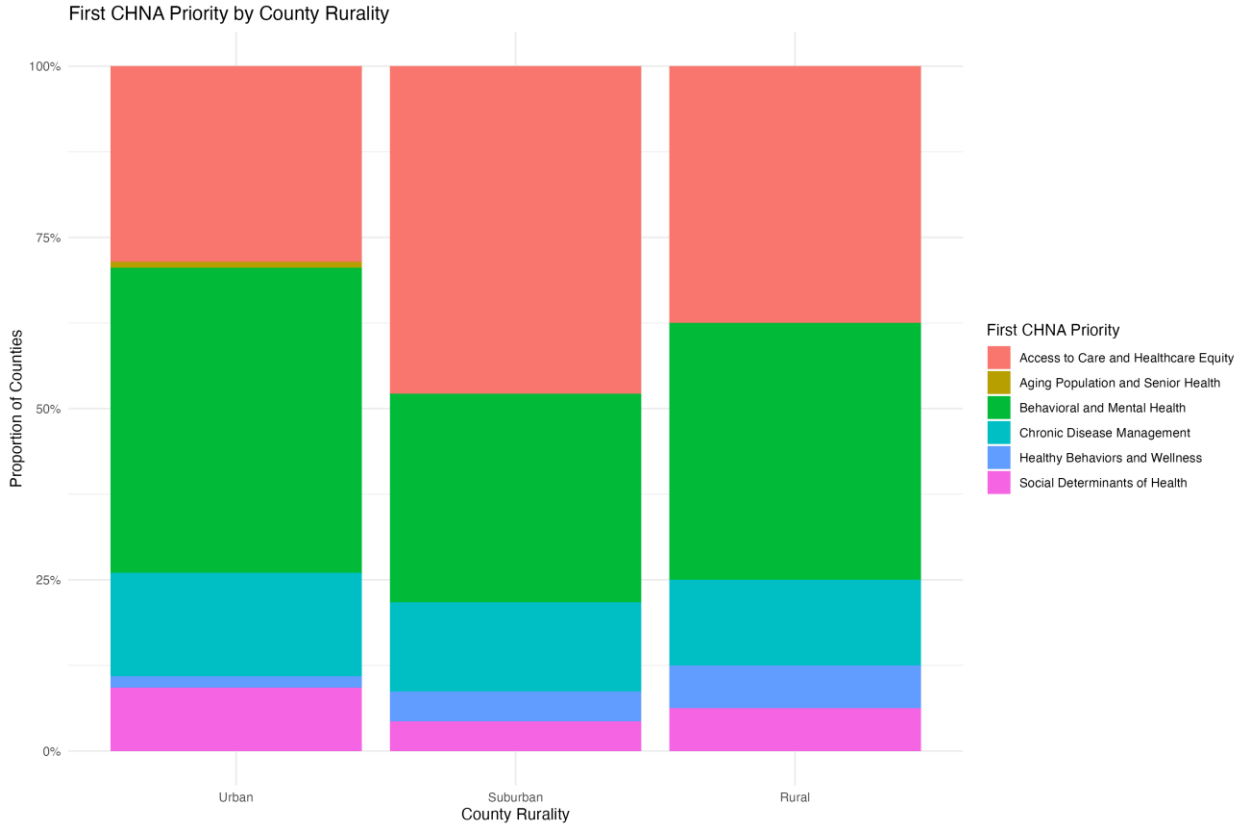
Priority overlap remains broadly distributed across counties of different population sizes, with the fitted line appearing nearly flat. This suggests little visible linear association between county population and the percentage of shared priorities between Community Health Needs Assessments and local health department IPLANs. Counties with similar populations show substantial variation in overlap, indicating that population size alone may not explain how closely hospital and local public health priorities align. The relatively small number of highly populated counties and the concentration of observations at lower population sizes should be considered when interpreting this pattern.

Distribution of Counties' First-Listed CHNA Priorities by Poverty Level



Access to Care and Healthcare Equity and Behavioral and Mental Health account for the largest proportions of first-listed CHNA priorities in both higher- and lower-poverty counties. Higher-poverty counties appear to have a larger proportion of Behavioral and Mental Health priorities, while lower-poverty counties show a larger proportion of Access to Care and Healthcare Equity priorities. Chronic Disease Management is also represented in both groups. These differences suggest variation in the priorities reported across poverty groups, although the figure alone does not establish statistically significant differences.

Distribution of Counties' First-Listed CHNA Priorities by Rurality Classification



Access to Care and Healthcare Equity and Behavioral and Mental Health are the most prominent first-listed CHNA priorities across urban, suburban, and rural counties. Access to Care and Healthcare Equity accounts for a particularly large proportion of suburban and rural counties, while Behavioral and Mental Health represents a substantial share of priorities across all three classifications. Chronic Disease Management is also consistently represented. The observed differences may reflect variation in community health needs and healthcare infrastructure, but additional analysis would be needed to determine whether the distributions differ meaningfully by rurality.

GIS Spatial Mapping Tool

Objective: Develop an interactive GIS mapping tool to spatially visualize and evaluate the geographic distribution of hospital Community Health Needs Assessments (CHNAs) alongside county IPLAN priorities across Illinois.

Application: Map hospital locations relative to county boundaries to visually identify regional health priority clusters, highlight areas of high versus low-priority overlap, and uncover spatial healthcare resource disparities (such as rural hospital deserts).

Impact: Provide public health leaders and researchers with a visual interface to better target cross-sector resource allocation and collaborative health interventions.

[View Interactive Illinois County Health Map]

